

When Play Hurts: Understanding Common Barriers in Movement-Based Games

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Figure 1: We conducted an online survey among 174 exergame players and identified common barriers encountered while playing, their effects, and players’ mitigation strategies. This figure shows the three popular exergames from our corpus. From left to right: Pokémon Go [76], played on a smartphone. Beat Saber [33], played with a VR headset. Ring Fit Adventure [80], played on a console.

Abstract

Exergames promise enjoyable physical activity through gameplay, yet players often face barriers that undermine engagement, safety, and retention. To date, knowledge about which barriers are encountered by end-users of commercial exergames and which mitigation strategies are used is limited. To address this gap, we conducted an online survey with 174 participants and provide a comprehensive organization of 60 reported barriers across six categories: physical, mental, social, environmental, technological, and game design. Key barriers include space limitations, social discomfort, addictive gameplay, and injuries. Our analysis reveals that while players try to mitigate barriers through ad-hoc strategies, issues like embarrassment, addiction, and harassment remain difficult to overcome. These findings highlight the need for more adaptive game designs, including dynamic spatial adjustments, personalized pacing mechanisms, and supportive social features. This work advances the understanding

of exergame barriers and their impact and offers actionable insights for designing more inclusive and resilient movement-based games.

CCS Concepts

• **Software and its engineering** → **Interactive games**; • **Human-centered computing** → *Empirical studies in HCI*.

Keywords

exergames, movement games, barriers, problems, games, survey

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1 Introduction

Virtual skiing using a Microsoft Kinect, gathering Pokémon during a daily walk around the neighborhood, or slashing cubes while dancing to music in virtual reality (VR) — these are all examples for so-called *exergames*. The term, a combination of “exertion” and “game”, describes a game genre that is played primarily through



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bodily, exerting movements [71]. Exergames date back to early dancing games like Dance Dance Revolution [6], and a consistent attention in games research (e.g., [37, 44, 69]). Apart from the inherent enjoyment, exergames have also been deemed having great potential to encourage people towards engaging in more physical activity [12, 99]. This potential makes exergames a timely research topic for multiple disciplines, such as medicine [99], movement science [98], and human-computer interaction [44], since it aligns with the world-health organization's goal often finding new ways of addressing the increasing physical inactivity [108].

Prior research has revealed various benefits of exergames [16, 46, 48, 109]. For example, training with an exergame has been found comparably enjoyable as using a personal trainer [64]. Other research revealed that immersive applications outperform traditional console exergames in terms of enjoyment and performance [7]. And exergames have been even used to train more complex, explosive movements like vertical jumps without risking injuries [20]. Outside of academic research, commercial exergames have experienced peaks of significant popularity: the Nintendo Wii [81] and Microsoft Kinect [68] brought sports games to many living rooms. Niantic's Pokémon Go [76] reached an impressive number of 250 million monthly users [36]. Lastly, exergames like Beat Saber [33] are also constantly among the best-selling genres of VR games.

However, this list of academic and commercial use also shows a typical trend: Exergame participation is often dwindling after the first engagement. While Pokémon Go still has a lot of users, it has only partially retained its first peak success [36]. Similarly, engagement is typically high during short-term or repeated experiments [65] but this enthusiasm often fails to translate into sustained use. For example, Rasche et al. [88] found that the average use of Pokémon Go is even shorter than of regular activity trackers. Naturally, retention in games dwindles after time. Nevertheless, exergames are meant to deliver lasting enjoyable experiences that encourage players to stay physically active. For this reason, it merits a closer look at what prevents players from fully enjoying them in the long-term. Is it lack of novelty, access to hardware, or do current exergame designs have barriers that discourage users from enjoying them fully?

While some researchers have explored the challenges of exergame design [88, 94], these publications primarily focused on individual games, platforms, or barriers. For example, the study by Shaw et al. [94] examined five self-identified design challenges, specific to VR headsets, using an exergame as a case study. The study by Rasche et al. [88] is possibly the one closest to our work: They surveyed 199 active and former Pokémon Go players on the motivational aspects of this game. While these works showcase important challenges in exergame design, they are not providing a holistic view across games, platforms, and user groups. To date, there has been no dedicated exploration of the problems that are experienced by end-users of commercial exergames during regular play. However, to unlock their potential as widely accessible tools against sedentary behavior, we must first understand and address the most pressing barriers, i.e., problems that prevent players from using and enjoying the exergames to their full intended extent.

With our research, we address this critical gap by comprehensively organizing barriers experienced by players of mainstream exergames. Identifying and structuring these common barriers is

essential for resolving them and thereby improving long-term engagement and accessibility. As first research, we therefore asked active and former exergame players for their encountered barriers. Apart from querying for the most common issues, we were also interested in their consequences on the play experience. Do players disengage entirely, or do they adapt their behavior to circumvent obstacles? Which strategies are players using to mitigate encountered barriers? To capture a broad spectrum of experiences, we included both current and former players and did not restrict the sample to currently played games. Additionally, we explored how the reported barriers differ between these device-classes to account for the big technological differences. As emphasized by previous studies [7, 30, 109], exergames differ greatly in design, theme, and interactions. In particular, exergames have been developed for many different types of hardware, including arcade machines, consoles (e.g., Xbox [67], Wii [81] or Switch [82]), and smartphones in the form of location-based games. In recent years, extended reality (XR) devices such as VR headsets and AR glasses have further expanded the exergaming landscape, offering new forms of immersive engagement.

Taken together, our four research questions are:

RQ1 What are common barriers encountered while playing commercial exergames?

RQ2 How do barriers in exergames differ between different technologies?

RQ3 What are the effects of encountered barriers on the exergaming experience?

RQ4 How do players overcome/mitigate encountered barriers?

To answer these questions, we created a qualitative survey to query which barriers are encountered by players. Therefore, we asked for encountered barriers across six categories—physical, mental, social, environmental, technological, and other—, the affected game, and the effect and mitigation strategy. We publicized the survey on the survey-platform Prolific¹ and gathered a final dataset of $n = 174$ responses, which we analyzed using a hybrid approach of thematic analysis that encompasses both elements of reflexive and codebook approaches [11].

At its core, our work delivers an empirical research contribution, according to Wobbrock and Kientz [107]'s classification of human-computer interaction (HCI) research contributions. The data collection happens via crowdsourced survey study and delivers three primary knowledge gains: First, we provide a descriptive overview of commonly played exergames and predominant platforms, revealing the huge impact early console games had on exergame players, especially when comparing them to more niche platforms like VR. Next, we provide a structured organization of prevalent barriers, including a cross-platform comparison and a differentiation between previously reported and newly surfaced issues. We also cover reported effects and mitigation strategies, showing that while players are well-versed in mitigating most issues on their own, frustration, embarrassment, and addiction are not easily circumvented and require the help of game designers, developers, and researchers. Based on these insights, we end our paper by discussing lessons learned for future exergame designs and how our five most interesting takeaways challenge established assumptions and call for

¹<https://www.prolific.com/>

more adaptive and personalized game designs. All three aspects of our work help clarify the shortcomings of currently available exergames, providing researchers with promising avenues for further studies and developers with actionable recommendations on how to make their titles more accessible and enjoyable for the general public.

2 Related Work

Sedentary lifestyles and a lack of physical activity are pressing problems that HCI scholars are designing innovative solutions for. In this context, exergames have gained significant traction as playful interventions for more physical activity, enjoyable rehabilitation method, or active social gaming sessions. In this section, we provide an overview of the current state of research regarding exergames' benefits (i.e., their "bright" side) and barriers (i.e., their "dark" side). This contrast helps contextualize our focus on barriers and their implications for long-term engagement.

2.1 Exergames

As technology evolved, so have games. We have advanced from traditional button- and keyboard-oriented games to experiences that involve the entire body and provide exercise opportunities through bodily interactions: *exergames*. These games require players to perform movements to play the games [71], for example, by hitting boxing targets in FitXR [32]. Over the years, exergames have been organized into many complementary frameworks; for an overview, see the work by Mueller et al. [70]. Historically, exergames gained popularity in the 1990s, with prominent examples like Dance Dance Revolution [6], which challenged players to perform stepping movements in sync with the music. Since then, rapid advances in technology have brought these games into our living rooms, for example with Wii consoles [81] featuring various games (e.g., Wii Fit [79]). Nowadays, users have even more options to experience these games. For example, they can use VR headsets (e.g., to play Beat Saber [33]) for a more immersive and enjoyable gaming experience [44], or ubiquitous technologies such as smartphones (e.g., to play Pokémon Go [76]) on their way to work or to meet friends [58]. Even first gyms have adopted immersive exergame technologies, like Black Box VR [105]. While these developments have broadened access and appeal, they also introduce new challenges in terms of usability, safety, and retention.

2.2 Bright Sides of Exergames

Exergame research shows that while these games are used for a multitude of purposes, their primary use case and biggest potential lie in combating insufficient physical activity [44]. Additionally, exergames can foster social interactions among users (e.g., [42]), which are an important motivator for continued exergame play [14]. For example, exergames can be thoughtfully designed to bridge generational gaps within families [49]. Such social engagement is particularly beneficial for older adults with dementia. For instance, Unbehaun et al. [103] introduced a Kinect-based music exergame specifically designed for this user group, which not only promoted physical activity but also facilitated social interactions. Similarly, a study on a VR exergame [47] revealed that these gaming experiences benefit the social engagement between older adults with

dementia and their caregivers. Wang et al. [106] explored different degrees of collaboration in collocated exergames, which possess the potential to foster teamwork and cohesion. Another popular sub-genre of exergames is location-based games like Pokémon Go [76] and Harry Potter: Wizard Unite [77], which motivate players to go outside and explore their surroundings [56] while also promoting physical activity [57]. Additionally, these games can foster connectedness to nature [93] and inspire players about environmental issues [54]. Arjoranta et al. [3] explored how game features of Pokémon Go affected different types of behavior change, for example by enriching daily routines or lowering social barriers.

Exergames are also used for schoolchildren [34] and have been explored as a fun way of engaging players with accessibility needs in physical exercise [35, 39]. Gao et al. [34] found casual exergames to be a promising tool to effectively decrease sedentary behavior for schoolchildren since they are preferred over traditional exercise. Hicks and Gerling [39] created a casual exergame that is designed for and controlled entirely by wheelchair movements to offer physical activity opportunities for children with wheelchairs. In an exergame study, targeting players with different skill levels, Gerling et al. [35] found that balancing approaches used in commercial games can reduce self-esteem. Instead, they advocate for hidden balancing that can increase the game experience for all users. Overall, research on the use of exergames revealed promising health-related results. For example, a systematic review by Li et al. [61] found that these games can help alleviate depression. Similarly, Sween et al. [99] reported that these games can achieve energy expenditure level expected from moderate-intensity exercises.

2.3 Dark Sides of Exergames

Despite their advantages, exergames are highly complex systems whose technical and design requirements create additional challenges for developers. From a technical perspective, these games require advanced tracking capabilities to detect users' movements accurately. Similarly, they must effectively integrate principles of movement design alongside game design to enhance the overall gameplay experience. The failure of any of these three components can have a detrimental impact on the gaming experience. Especially, movement design-related failures might negatively affect players' health. For example, an interview with sports science experts by Cmentowski et al. [20] surfaced their concerns regarding unsuitable exercises and training intensities in the context of jump training. Moreover, retention in both games² and sports [29] is notoriously low, raising questions about the long-term impact of exergames beyond initial novelty.

An increasing amount of literature has focused specifically on challenges and motivational barriers for individual games or user groups. For instance, Rasche et al. [88] explored barriers encountered by Pokémon Go players and listed required improvements of in-game features, such as offering more challenges. Similarly, Adam and Senner [1] surveyed 54 active or former young- and middle-aged exergame players using an online questionnaire and found that fun, affiliation, and competition are important motives for playing, whereas unmet expectations towards health or losing weight benefits were main reasons for quitting playing among

²<https://www.businessofapps.com/data/mobile-game-retention-rates/>

mainly younger players. Unfortunately, it remains unclear which games the participants played. Kaos et al. [43] found that a feeling of program belonging kept group players engaged in exergames; in fact, playing multiplayer exergames alone led to worse adherence than for pure singleplayer games. As one of the few works trying to mitigate barriers, Shaw et al. [94] identified five design challenges of VR exergames (e.g., cybersickness and motion tracking) and developed a VR game to explore their resolution.

Apart from these more general investigations, many studies focused specifically on user groups with additional accessibility and support requirements: In Barg-Walkow et al. [5]’s work, older adults engaged with two Kinect-based programs. While participants were generally satisfied with using the exergames, they believed the system was overly complex and emphasized the need for technical assistance. Similarly, Harrington et al. [38] found mainly usability challenges experienced by older adults while engaging with Kinect games. In contrast, Crane et al. [23] pointed out frustration resulting from unfair level design, the inability to complete levels, fatigue, pain, and problems with the camera tracking. A review by Ravani et al. [89] involving patients with heart failure concluded that users face barriers associated with boredom, personal preferences, and technical constraints. Chao et al. [15]’s study with assisted living residents additionally reveal age and health related problems that make exergaming a difficult and unpleasant experience, including fear, tiredness, and frustration. Interestingly, while some prior literature suggests that older adults value health benefits more than in-game challenges [96], recent work suggests that in-game competition remains a critical aspect for older adults’ game enjoyment [45].

In summary, prior research has identified a range of barriers, but these are often studied in isolation or within specific user groups. A comprehensive cross-game, cross-platform overview of the issues encountered by regular end-users is still missing. We argue that a survey of challenges in everyday exergame use is both timely and necessary to complement existing lab-based and targeted studies. Given that the production of exergames continues to rise, addressing these barriers is crucial, since neglecting them could negatively impact users’ health as well as the commercial success of these types of games. Our work will advance the research domain by replacing the guesswork and assumptions with a quantified overview of actual experienced problems, enabling further focused research and goal-oriented improvements that better meet the actual needs of end-users.

3 Survey

To investigate our research questions, we decided to conduct a qualitative online survey. The main reason for choosing this methodology is its ability to query a large and diverse group of participants from around the world without geographical constraints [59]. Surveys are useful for providing “insights into users’ [...] experiences” when “needing to represent an entire population” [72]. They allow a good overview of the general perceptions within the targeted population. However, it comes at the price of a lower information and contextual depth compared to alternatives such as semi-structured interviews [59]. This methodology best suits our goal of prioritizing breadth over depth and identifying frequent barriers across a

diverse exergaming population. Given that prior studies already explored individual barriers and games in smaller user studies and through detailed interviews and that our goal was to provide a comprehensive overview that does not focus on the experiences or opinions of few individuals but comprehensively organizes global patterns and trends, we further decided to *only* rely on the survey data for analysis. At the same time, prior work has not yet assembled a complete list but instead mostly discussed individual barriers. Therefore, we decided to keep the survey as open as possible to capture both known and new barriers. As distribution platform for the survey, we chose the online recruitment site Prolific³, similar to previous HCI studies [40, 54]. While there are additional risks associated with using online recruitment platforms, such as self-selection biases, their suitability for data collection has been confirmed by previous research [27, 31]. In particular, Prolific has been found to yield higher data quality than comparable competitors [87]. To ensure data quality, we incorporated attention check questions [84] in our survey and decided on the following screening criteria within Prolific: only users who actively play video games and are fluent in English were deemed eligible to participate.

3.1 Questionnaire Design

Our survey targeted players who are playing or have played exergames, regardless of technology or game. The first part started by querying basic demographic information, i.e., age, gender, exergame experience, and all exergames that participants have played or are playing. Since the understanding of what makes a game an exergame might be challenging (e.g., required movement to be considered, see [44]), we provided a working definition and representative examples across platforms to participants to ensure consistent interpretation. For the working definition, we simplified Mueller et al. [71]’s understanding that exergames are primarily played through bodily, exerting movements by just defining them as “any game played mainly with body movements.” Using simpler language, we aimed to reduce possible barriers of participation. Additionally, we listed seven example games to help with understanding: Dance Dance Revolution, Pokémon Go, Beat Saber, Wii Fit, Kinect Sports, FitXR, Harry Potter: Wizards Unite. We chose the most renown games (i.e., most played and/or most media coverage) for each available technology (arcade, smartphones, VR headsets, consoles) to minimize potential anchoring effects. Even with this information, we had to exclude few participants because they misinterpreted games like FIFA as exergame. After the demographics and gaming habits, the second survey part consisted out of six tables asking about barriers. Taken together, our survey was structured as follows:

- **Demographics:** In this section, players were asked about their gender, age, and years of exergame experience.
- **Exergames:** This section asked players to list any exergame they have played or are playing, the device they used to play that exergame, and how often they played it.
- **Barriers:** In this part of the survey, we inquired about the barriers players faced when playing an exergame (see below).

³<https://www.prolific.com/>

Physical Barriers

Please answer questions about physical barriers you have experienced while playing exergames.

Physical barriers: All barriers you encountered that relate to physical challenges, e.g., injuries, exhaustion, overtraining, etc.

	In which exergame did this barrier occur?	Which physical barrier do/did you experience while playing this exergame?	How often do/did you experience this issue while playing this exergame?	What is/was the effect on you and how do/did you overcome/mitigate it?
#1	"Ring Fit Adventure" (P77)	"hurt my knee recently" (P77)	"once a week" (P77)	"wore a knee brace" (P77)
#2				
#3				
#4				
#5				

Figure 2: All the barrier categories had the same questions. Here, we provide an example from the physical barrier category for illustration purposes and provide an example answer from one of our participants.

Categories of Barriers. The core of our research focuses on the barriers encountered while playing exergames. Therefore, we wanted to get a more granular understanding of the barriers players experience. To capture the nuanced perspective of players, we created in-depth categories of barriers that can occur while playing these games. To formulate and create our categories, we started by collecting barriers mentioned in prior literature. Then, we clustered these notions into distinct categories using affinity mapping. Discussing among the first two authors, both experienced in exergame research, we reclustered and refined the categories in multiple iterations to ensure clear distinction between them. Many papers discussed health-related issues, such as age-related impairments [15], fatigue and pain [23], or motion sickness [94]. We grouped these into the category of *physical barriers* originating from the players themselves. Shaw et al. [94] also discussed challenges originating from the surrounding *environment*, such as the risk of colliding with the furniture. One of the most common type of barrier, raised by all considered publications, are *technical* issues. This includes motion tracking [23], server connections [89], and latency [94]. In particular Rasche et al. [88] focused more on players' feelings and emotions: boredom, disappointment, and interest to play. Together with unmet expectations [89] and frustration [15], we grouped those factors into *mental barriers*. Lastly, Ravani et al. [89] discussed the wish to play together with other people as a reason to quit. Our fifth category, *social barriers*, incorporates such aspects of social play (e.g., harassment, embarrassment, toxicity [52]). To account for barriers not fitting existing categories, we included an *other barriers* category, which was later refined into *game design barriers* during the thematic analysis. Concluding, our six categories are:

- **Physical barriers:** All barriers players encounter that relate to physical challenges, e.g., fatigue, pain, and injuries..
- **Mental barriers:** All barriers players encounter that relate to mental challenges, e.g., boredom, frustration, unmet expectations, and addiction.
- **Social barriers:** All barriers players encounter that relate to social challenges, e.g., isolation, arguments, harassment, and embarrassment.
- **Environmental barriers:** All barriers players encounter that relate to environmental challenges, e.g., collisions, space, and pets.
- **Technological barriers:** All barriers players encounter that relate to technological challenges, e.g., bugs, hardware-failures, GPS, device support, and privacy.
- **Other barriers:** All barriers players encounter that relate to other challenges.

For each category of barrier, the survey provided a table, querying which barrier, in which game, and how often it occurred. In addition, we asked about the impact of this barrier and participants' strategies to overcome it. This open-ended design was chosen to capture occurrences of both established and unknown barriers while also reducing the risk of satisficing and acquiescence [72]. Please refer to Figure 2 for an example of how participants experienced the survey for the six categories. In total, our survey took on average 11 minutes to complete. For their efforts, participants were reimbursed with [anonymous for review], according to Prolic's regulations as paid data gathering platform. The full list of survey items is available in the supplementary materials to support transparency and reproducibility.

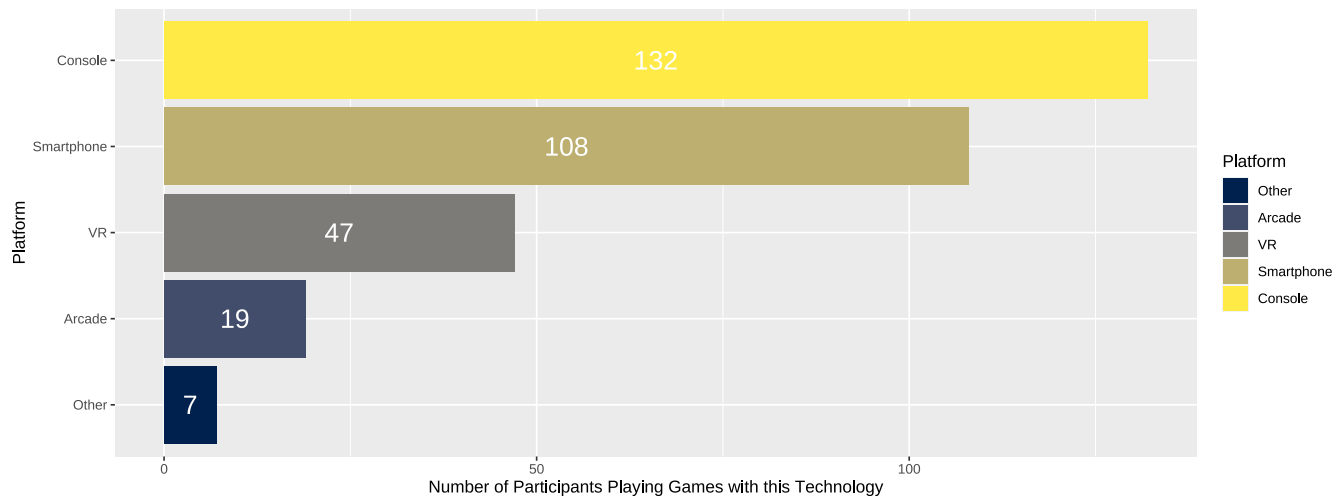


Figure 3: The number of participants playing games on each platform: Console exergames are the most played exergames, followed by smartphone games. VR, arcade, and other titles are much less prevalent.

4 Analysis

Our survey collected qualitative data as well as demographic information. Using descriptive statistics, we report demographic information and support reporting our qualitative data. We analyzed our qualitative data employing a hybrid approach that combines primarily a codebook approach (e.g., involving multiple coders in the several points of the process, providing topic summaries) with the reflexive orientation (e.g., reflection, in-depth interpretation) of thematic analysis [10, 11, 19]. While codebook approaches offer a more structured process for coding and present pragmatic advantages (e.g., efficiently manage large datasets), they do not aim for consensus among coders, unlike coding reliability approaches of thematic analysis [11]. Importantly, since our focus was on exploring the depth and richness of the data rather than achieving coder consensus, we did not treat researcher subjectivity as a bias. Instead, we viewed it as an asset that enriched our interpretation of exergame-specific insights, aligning with the perspective of Clarke and Braun [19]. Consequently, unlike coding reliability approaches [11], we did not assess inter-coder agreement and focused instead on a nuanced, interpretive understanding of the data.

For the coding process, we used the online platform Dovetail⁴ due to its flexibility and GDPR-conform data storage. We did not enable or use any AI features for our analysis. Since Prolific already pseudonymizes responses, we could directly upload the data after removing incomplete entries. Then, two authors created deductive codes for each barrier category (e.g., physical barrier → injury, technological barrier → bugs). Then, both authors independently coded the first batch (10%, 21 responses) using these deductive codes. However, they also created inductive codes when necessary. After this first batch, they calibrated their codes, including renaming, merging, or creating codes, in an online meeting. This meeting involved both conflict identification and uncovering the meaning of data perceived differently by the authors. Afterwards, the lead

author coded the remaining data on their own using this codebook while adding inductive codes when necessary. Difficulties during the individual coding process were discussed between both authors. After the coding, the lead author again met with the second author to review the resulting codebook. Finally, both authors drafted the initial themes together and refined these by matching them to the participants' quotes. For the final frequency calculation, we used R version 4.5.1. Please see the supplementary materials for the codebook and code frequency.

Following best practices [19, 60, 74, 83], both authors who are involved in qualitative data analysis present a statement of reflexivity: Both authors have been conducting exergame research and qualitative data analysis for several years, including their Ph.D. studies. They have a positive stance on exergames and regard these games as an innovative solution to address insufficient physical activity. They engage with exergames themselves to stay informed about current innovations and to incorporate physical activity into their routines. However, in this research, they challenge their own research perspective and focus on understanding the barriers of exergame applications with the goal of better understanding the needs and experiences of users. Their personal experience with playing exergames enables them to relate to the barriers, and their technical background allows for an in-depth analysis of technical and design aspects. Nevertheless, we note that they lack expertise in movement science and medicine, which could help them better understand the physical and mental barriers faced by players. We also reflect on our methodological choice: We opted for an online survey to reach a broader participant base compared to the other methods (e.g., interviews). We consider the online survey a suitable initial step for obtaining a comprehensive overview of the barriers associated with exergames, while interviews might be beneficial for exploring individual games or conducting in-depth interviews to examine detailed or context-specific barriers.

⁴<https://dovetail.com/>

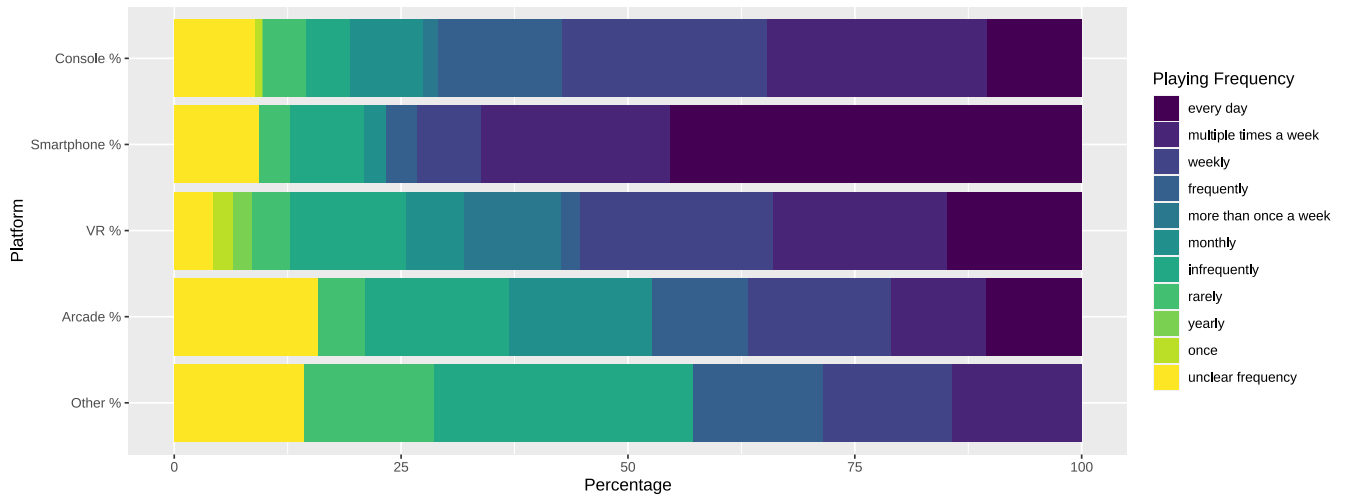


Figure 4: The gaming habits of participants per device platform: from top to bottom, the rows show the frequency with which participants play smartphone, console, VR, arcade, and other games. Smartphone exergames are played most frequently.

5 Results

In this section, we present the results from both the descriptive analysis of demographic data and the qualitative data analysis. The latter encompasses three parts: first, we provide an overview of the played exergames, used devices, and playing frequencies. Next, we summarize the identified barriers separated by categories. In our thematic grouping, we identified an additional category *game design barriers* that replaces the original *other barriers* category. For each occurrence of all barriers, we aimed to identify the precise barrier, i.e., the underlying problem, and the affected exergame, but also the *effect* on the experiencing participant and their *mitigation* strategy, if applicable. While we also queried for the frequency with which the barrier occurs, our data mostly consists of very broad terms like frequently or infrequently. We do not see much value in reporting this data as any barrier—regardless of how often it occurs—can have a negative consequence, such as a player abandoning a game. Therefore, we focus on the barriers, effects, and mitigation. In total, all participants reported over 600 problem statements that were grouped into 60 distinct barriers (for a full overview, see supplemental material), 44 effects, and 53 mitigation strategies.

5.1 Demographics

We received complete responses to our survey from 205 participants. Although they completed the entire questionnaire, we had to exclude some due to nonsensical responses. As exclusion criterion, we checked if the reported games existed and if the used devices matched, for example, Pokémon Go is not available for PC and Wii Fit not for VR. We aimed to be as inclusive as possible and discussed every case among the team. However, we still identified 31 entries with clear fraudulent entries. For example, participants claimed playing “sport simulation” on “device” or “Beat Saber” on “Accelerometer”. In two cases, responses matched almost word-by-word, despite Prolific’s integrity measures. After removing these

responses from our dataset⁵, we ended up with a final dataset of 174 complete responses. Our sample was slightly skewed in gender: men=96, women=74, non-binary=4. The average age was 29.33 ($SD=8.17$, $Min-Max=18-62$) and participants played exergames for on average 5.64 years ($SD=5.40$, $Min-Max=a\text{ week}-22\text{ years}$).

5.2 Exergames and Devices

Overall, we identified 64 exergames played by our participants (for a full overview, see supplemental materials). The most common ones include Pokémon Go ($n=105$), Just Dance [102] ($n=57$), and Wii Fit ($n=47$). These and the other games were played on 14 different devices, which we clustered into 5 platforms: game consoles, smartphones, VR headsets, arcade machines, and other devices (see Figure 3). 132 participants reported playing console games (Nintendo Wii: 87, Nintendo Switch: 35, Microsoft Xbox: 35, Sony PlayStation: 29). The top-3 console games were accordingly Just Dance ($n=55$), Wii Fit ($n=47$), and Wii Sports [78] ($n=45$). Second in place were location-based games played using a smartphone ($n=108$). Almost everyone played Pokémon Go ($n=105$), with only a few people trying alternatives like Zombies Run ($n=7$) or Harry Potter: Wizards Unite ($n=6$). Despite being praised for their suitability for exergames, VR headsets have so far only attracted 47 participants (mobile headsets: 16, PC-bound headsets: 10, PlayStation VR: 9) who primarily played Beat Saber ($n=39$). Only very few participants played other games like Supernatural ($n=4$) or FitXR ($n=2$). Exergames started in arcade halls; nevertheless, only 19 participants reported experience using arcade machines, with all playing Dance Dance Revolution ($n=19$) and few also Pump It Up! ($n=2$) and Dancerush Stardom ($n=1$). Lastly, we grouped rarely-used devices into a miscellaneous category ($n=7$), encompassing the PC

⁵Comparing the excluded 31 fraudulent entries to the final sample of 174 responses, the demographic is similar regarding age ($M=28.35$, $SD=8.56$) but excluded participants had less exergame experience ($M=3.98$, $SD=3.52$) and showed an opposite gender imbalance: men=9, women=22.

($n=5$) for playing Step Mania ($n=2$), an interactive wall called TWall ($n=1$), and Apple Watch-based smartwatch games ($n=1$).

We also asked participants for their gaming habits when playing or having played exergames (see Figure 4). For the smartphone games, most people reported playing every day ($n=52$) or multiple times per week ($n=20$). Although more people had experience with console exergames, their play frequency was less: most participants played them one ($n=29$) or more times per week ($n=33$), with only 14 people playing daily. VR games show a similar play pattern: most played them once ($n=10$) or more times ($n=9$) per week, with less players playing daily ($n=7$). Arcade machines are rarely located at home, which might explain the lower frequency: 3 participants each play arcade games weekly, monthly, or infrequently. Lastly, devices in the "other" category are played infrequently ($n=2$) or rarely ($n=1$). While our focus is on general use frequency in present and past, some participants explicitly reported having stopped playing: 17% of console players (21), 9.3% of smartphone players (8), 26% of arcade players (5), and 4.2% of VR players.

5.3 Environmental Barriers

The largest category of barriers concerns ● *environmental* factors and incorporates 152 reports of 9 issues (see Table 1). The 3 most common barriers are limited space, collisions, and geological limitations. While causing safety issues and frustration, environmental barriers can be mitigated through careful setup or safeguarding measures.

Most participants have problems with limited space ($n=61$). While this is not an issue with outdoor location-based experiences, games like Just Dance ($n=16$), Wii Sports ($n=11$), or Beat Saber ($n=6$) typically require an obstacle-free medium-sized space in the living room. If the space is too small, people experience reduced performance ($n=5$), safety risks ($n=2$), or even not being able to play at all ($n=2$). Yet, these effects are generally rare, and most participants were able to easily resolve this barrier by making available space ($n=27$), e.g., by rearranging furniture, or by changing location ($n=13$), e.g., to another room. Despite these modifications, participants still regularly experienced collisions with the environment ($n=28$), especially when playing Beat Saber ($n=7$), Just Dance ($n=6$), and Wii Sports ($n=4$). Reported effects include damage to surroundings ($n=5$)—"Luckily no expensive damage so only scratches to the VR controllers and the objects I hit (the wall)." (P8)—, obtaining injuries ($n=4$), or feeling at risk ($n=2$). Since collisions are closely linked to limited space, they can also be resolved by creating bigger play areas ($n=8$) and securing the environment ($n=3$), or by simply being more careful ($n=4$). Multiplayer games like Just Dance ($n=4$), Wii Sports ($n=2$), or Dance Dance Revolution ($n=1$) additionally bring the risk of collisions with other players ($n=7$). Interestingly, participants did not report any effects from this barrier. The only way to prevent such collisions seems to be an increase in available play space ($n=5$). In contrast, collisions with pets and bystanders seem to be a much bigger issue ($n=16$). Across games, these were reported as safety risks ($n=3$) and interruptions ($n=1$) which can lead to lost progress ($n=1$). As far as possible, participants therefore attempted

to secure the environment prior to playing ($n=5$), e.g., by locking pets out of the room.

When playing location-based smartphone games like Pokémon Go, participants experienced an entirely different range of barriers. Geographical limitations ($n=16$) occur when public locations are not accessible due to limited entry, inaccessible terrains, or other blockades: "I was living in a small city near a mountain and there were spots in the game which I could not reach in real life" (P109). When experiencing such issues, participants felt frustrated or sad, but also feared safety risks (each $n=1$). Therefore, most of them moved to other locations ($n=7$) or simply avoided the affected content, such as the Pokémons that are unreachable ($n=6$). Another problem when playing outside is bad weather ($n=9$), forcing participants to play less ($n=2$) to avoid getting wet ($n=1$) or possibly damaging the electronic device ($n=1$). Similarly, playing in a public space typically introduces distractions from the surroundings ($n=8$). With noise, participants may opt for headphones ($n=2$); otherwise, they would just change locations ($n=4$). Lastly, few participants experienced other problems relating to external events, such as curfews, blackouts ($n=3$), and the play area (e.g., slippery floor, bad lighting, heat, $n=4$).

5.4 Physical Barriers

Since exergames are predominantly played by bodily movements, ● *physical barriers* form another important aspect of the experience (10 issues, 138 reports, see Table 1). The 3 most common barriers are the natural exhaustion from participating in movement-intensive games, injuries, and general fatigue. Since lesser issues in this category can accumulate, proper pacing and regular breaks are essential mitigation strategies.

Interestingly, the most frequently stated physical barrier is not necessarily a "problem", but the inherent nature of *exertion* games: participating in movement-intensive gameplay ($n=37$). After playing these games, participants naturally feel exhausted ($n=37$) and accordingly take breaks ($n=18$), rest ($n=5$), and hydrate properly ($n=3$). However, physical exercise can also lead to more serious concerns. Many participants reported cases of fatigue and muscle soreness across all games, requiring them to take longer breaks ($n=7$) and stretch ($n=4$), such as "making pauses to stretch between different songs" (P112). Since participation can even lead to overstraining ($n=16$) without precautions measures, some try to be more careful ($n=4$) and limit their playtime ($n=4$). Another severe consequence of engaging in exergames is injuries ($n=25$), including broken legs or sprained ankles. In our sample, most injuries were reported with Just Dance ($n=8$), Pokémon Go ($n=4$), Ring Fit Adventure and Beat Saber (each $n=3$). This did not only force participants to stop playing for a while ($n=4$) but also affected their other obligations ($n=1$). Apart from seeking medical help ($n=3$), they tried to be more careful ($n=3$), create more space ($n=3$), and use proper warm-ups/cooldowns and protective supports ($n=\text{each } 2$), e.g., wrist supports, to prevent further injuries.

Other reported issues were connected more to playing styles or specific devices. Notably, our data contains 20 reports of pain, e.g., wrist pain from repetitive movements in Beat Saber ($n=8$) or ankle

Table 1: The 31 reported barriers in the environmental, physical, and technological categories. The index numbers indicates their prevalence. The mentions column lists previous work that mentioned the barriers as motivation, discussion point, or potential issue. The findings/solutions column lists study results or developed solutions that help understand or resolve the barriers.

BARRIERS	MENTIONS	FINDINGS/SOLUTIONS
● Environmental Barriers		
#1 limited space ($n=61$)	[75]	-
#5 collision ($n=28$)	[4, 17, 22, 73, 94]	-
#12 geological limitations ($n=16$)	[4, 8, 17, 58, 88]	-
#12 pets and people ($n=16$)	[18]	pet-detection prototype [18] to increase safety
#15 weather ($n=9$)	[63]	weather significantly affects walking distance in Pokémon Go [63]
#16 distracting surroundings ($n=8$)	[104]	-
#17 collision with other players ($n=7$)	[101]	players sharing play area fear collisions and have lower team involvement in VR [101]
#20 problems with playspace ($n=4$)	[20, 104]	-
#21 external events (e.g., curfews) ($n=3$)	-	-
● Physical Barriers		
#2 playing movement-intensive game ($n=37$)	[5, 15, 70]	-
#7 injury ($n=25$)	[20, 70]	-
#8 fatigue ($n=22$)	[23, 75]	-
#9 pain ($n=20$)	[23]	-
#12 overstraining ($n=16$)	[20, 70, 92]	nudges effective for preventing overexertion [92]
#16 extended use ($n=8$)	[23, 104]	especially for older adults, therapists need to monitor and schedule breaks [23]
#19 VR issues (e.g., cybersickness) ($n=5$)	[20, 45, 94]	matching virtual and real world prevented cybersickness in a jumping exergame [20]
#19 dizziness/balance problems ($n=5$)	[20, 94]	
#21 lacking fitness ($n=3$)	[5, 20, 75]	-
#22 dehydration and overheat ($n=2$)	[73]	-
● Technological Barriers		
#6 hardware-failures ($n=27$)	[89, 104]	-
#9 connection (internet) ($n=20$)	[89]	-
#10 tracking accuracy/problems ($n=18$)	[20, 23, 62, 94]	tracking errors lead to frustration and require modifications [23], lack of tracking precision allows for compensatory movements [75]
#13 bugs ($n=15$)	[73, 88]	glitchy AR features impede exergaming [73]
#13 GPS ($n=15$)	[8]	-
#14 device support ($n=11$)	[73]	older phones prevent AR use [73]
#18 lag/delays ($n=6$)	[13, 41, 94]	-
#18 battery drain ($n=6$)	[73]	AR features not used due to battery drain [73]
#18 cable ($n=6$)	[20, 45, 94]	-
#18 availability (setup, maintenance) ($n=6$)	[47, 62, 70]	-
#19 ergonomics ($n=5$)	[20, 45]	discomfort from VR reduces immersion [45]
#22 graphics ($n=2$)	-	-

pain from jumping in Zumba Fitness ($n=1$). This causes discomfort ($n=3$) and forces participants to play less ($n=2$). To avoid the pain, they report taking more frequent breaks and rests ($n=10$) and changing their playstyle ($n=3$), e.g., adopting a different controller grip. Similarly, some participants reported feelings of exhaustion ($n=8$) and tiredness ($n=2$) after long play sessions ($n=8$). 5 reports link specifically to VR-related issues, i.e., cybersickness symptoms and eye strain. Participants playing Beat Saber ($n=4$) and Wizard VR ($n=2$) therefore took breaks throughout the experience ($n=3$)

or tried adjusting the headset's settings ($n=3$). Closely linked to cybersickness are dizziness and balance problems ($n=5$). Nevertheless, these not only occur with Beat Saber ($n=2$) but also console games like Wii Fit and Kinect Sports ($n=1$). Affected participants have to stop playing ($n=1$) and sometimes dabble with the resulting impediments for a longer time ($n=1$). Unfortunately, there is no real mitigation except for taking breaks ($n=1$), though clearing more space can at least give a better feeling of safety ($n=1$). Few participants also commented about the troubles starting a physical activity

after being sedentary ($n=3$), which can easily exhaust ($n=2$) and overwhelm ($n=1$) them—effects that luckily disappear with more practice and improved fitness ($n=2$). Lastly, two players experienced dehydration after playing for extended durations ($n=1$) and in the summer sun ($n=1$), highlighting the importance of proper breaks and hydration ($n=2$).

5.5 Technological Barriers

Technological devices are central to experiencing exergames, and so are ● problems with these. Overall, we collected 130 reports across 12 issues (see Table 1). The 3 most common barriers are hardware failures, internet connection problems, and tracking issues. Since many of these barriers are out of players' control, they typically cause major frustration and lead to disengagement.

The most predominant technological barrier is a hardware failure ($n=27$). When the phone overheats while playing Pokémon Go ($n=4$) or the balance board for Wii Fit ($n=4$) is not detected, this interrupts the gameplay ($n=4$), leading to frustration ($n=3$) and even permanent disengagement ($n=2$). Depending on the problem, players can only try to fix the problem ($n=6$) or have to upgrade/replace the device ($n=8$). Less permanent are problems connecting to the internet ($n=20$), although they are a source of annoyance ($n=2$) and can lead to loss of progress ($n=2$), for example, when playing Pokémon Go ($n=10$) outside. Closely connected are GPS problems ($n=15$), which are a barrier only observed for location-based smartphone games. Frustrated players can only try to restart the application ($n=4$) or change their location ($n=3$). The counterpart for VR and console games are tracking accuracy problems of the required controllers or motion trackers ($n=18$). Players recalibrate the sensors ($n=6$), adjust the settings, or simply ignore the problem ($n=4$).

Other common problems were lags and delays ($n=6$), for example, in Beat Saber ($n=2$) or Zumba Fitness ($n=2$). Since these can lead to missed movements, they generally decrease enjoyment ($n=1$). Participants tried to decrease the likelihood by adjusting the settings ($n=3$), e.g., reducing the graphical quality. Apart from these specific issues, several participants also reported bugs in general ($n=15$), which, if not mitigated by restarting ($n=5$) or updating the application ($n=2$), can lead to them quitting the game entirely ($n=2$). Despite being frustrating, bugs do not compare to an end of device support ($n=11$). Participants experienced both that Pokémon Go ($n=5$) was no longer playable on their smartphone and that Microsoft discontinued support for the Kinect, shutting them out of Just Dance ($n=2$). Since the only mitigation is to upgrade the device ($n=3$), players are instead likely to stop playing completely ($n=2$). Sometimes, the game is just not available temporarily, when requiring maintenance ($n=6$). This barrier is mostly observed with Dance Dance Revolution ($n=2$) and interrupts both the gameplay ($n=2$) and motivation to play ($n=1$).

The last group of technological barriers concerns specific platforms. Smartphone-based games particularly suffer from battery drain ($n=6$), which can be circumvented by buying battery packs ($n=3$). In contrast, VR users suffer from the necessity of cables ($n=6$)

which annoy ($n=3$) and require them to be more careful ($n=2$). Another issue is bad ergonomics of the headsets ($n=5$), leading to discomfort, physical limitations, or even quitting (all $n=1$). Lastly, two participants complained about bad graphical quality of the VR headset in Beat Saber ($n=2$), Pistol Whip ($n=1$), and Acorn Attack of the Squirrels ($n=1$). This decreases immersion ($n=1$), motivating one player to upgrade their device ($n=1$).

5.6 Mental Barriers

While ● mental barriers are less pronounced compared to the first three categories (9 issues, 88 reports, see Table 2), their effects are much more severe. The 3 most common barriers are addiction, in-game failures, and lack of motivation. Such concerns directly impact users' emotional state and personal life, with mitigation requiring careful adaptation of gaming behavior.

The most common problem in this category, according to participants, is becoming addicted to the game ($n=30$). Across platforms, games like Pokémon Go ($n=16$), Just Dance ($n=3$), and Beat Saber ($n=2$) “trapped” players into an addicting gameplay loop, affecting their personal life and other obligations ($n=6$). If temporary breaks ($n=6$) or playtime limitations ($n=3$) did not help, some participants had to quit the game entirely ($n=5$). However, some addictions also seem to be triggered by external factors in the player's life: “*Sometimes when other things in life weren't really going that well, I focused too much on the game*” (P45). On the opposite side, in-game failures ($n=27$) lead players to feel discouraged ($n=4$) and doubting themselves ($n=3$). To mitigate these feelings, they adjusted the difficulty ($n=6$) or focused on their personal progress ($n=3$). Furthermore, some participants experienced a general lack of motivation and commitment ($n=8$) to partake in exergaming, leading to less play ($n=3$). As solutions, they tried setting achievable goals ($n=2$) or stopping playing for a while ($n=2$).

Players of location-based games like Pokémon Go ($n=6$) and Monster Hunter Now ($n=1$) might get so cognitively involved ($n=6$) that they forget their surroundings, leading to safety risks ($n=4$). Some paid attention to stay more aware of the environment ($n=3$), while others preferred to change location ($n=2$) to be able to get immersed without negative consequences. When playing exergames, users typically have expectations of goals. If these are not met ($n=6$), this can lead to frustration and disappointment ($n=2$), which happens most often for VR exergames like Beat Saber ($n=2$) and Audica ($n=1$). Setting achievable goals ($n=1$) and practicing more ($n=1$) can help mitigate these feelings. Other participants are not discouraged but rather overly perfectionistic and competitive ($n=5$). Since this ultimately reduces the fun part of the experience, these players try to actively focus on the fun aspect ($n=4$) and thereby find peace with themselves ($n=1$). Few participants even went deeper into the loop of negative mood and experienced self-devaluation ($n=4$) and overthinking ($n=2$). In these cases, it might be better for the mental health to quit the game ($n=2$) or at least take breaks ($n=1$). Finally, one interesting case was about collision anxiety [91] in Beat Saber ($n=1$). This participant would only play with their partner present

since this gave them a feeling of security: “*I need my boyfriend to be there and warn me*” (P65).

5.7 Social Barriers

● Social barriers, reported 83 times, encompass 11 problems that arise from the interplay between the exergame player and other people, both within the game and outside of it (see Table 2). The 3 most common barriers are playing the game in-front of other people, experiencing isolation, and arguments with other players. Mitigation strategies include building confidence through practice and building like-minded communities.

The predominant issue that most participants struggled with was playing the game in front of others ($n=36$). This concerns mostly games with expressive movements, like Just Dance ($n=13$), but also other titles are affected, like Pokémon Go ($n=7$) or Beat Saber ($n=2$). Participants state feeling embarrassed ($n=31$) or fearing being judged ($n=5$): “*I was afraid I could be seen by people and notice I was playing the game, which felt weird as I’m already an adult*” (P83). While some quit ($n=6$), most players build confidence by first playing in private ($n=9$) or with friends ($n=5$). At the same time, the opposite is also true: some participants experienced isolation ($n=11$) from playing alone, causing them to feel lonely ($n=2$). If possible, these players try to actively search for like-minded peers to connect over the game and play together ($n=2$). Unfortunately, this attempt can sometimes be fugitive, since participants also reported a lack of players ($n=8$), for example, when playing Pokémon Go ($n=4$) in areas without many players of their age group. Generally, peers are not only needed for multiplayer modes; instead, participants also lament a lack of social support ($n=3$), which decreases their motivation to play ($n=1$).

Integrating exergaming into everyday life can sometimes be challenging. Seven participants noted experiences where their gaming activity clashed with other obligations, for example: “*When I played it, I had no time to attend my children’s needs*” (P1). To avoid affecting their personal life ($n=1$), participants were either required ($n=2$) or decided voluntarily ($n=2$) to limit their playtime. Similarly, some participants were worried to affect other people in their vicinity ($n=4$), e.g., by disturbing their sleep. Accordingly, they opted to play at different times ($n=2$) or change their play style ($n=1$). When playing together, words might also get heated and lead to arguments ($n=9$). This, of course, primarily concerns games typically played with multiple players, like Dance Dance Revolution ($n=2$) or Wii Sports ($n=2$). Participants would then pause the game ($n=2$) and try to resolve the argument ($n=2$). A much more extreme social barrier is harassment ($n=3$), which was sadly experienced, for example, in Pokémon Go ($n=1$). Affected players felt unsafe, distressed, and even horrible (each $n=1$). Even if intentions are not malicious, even slight teasing ($n=1$) of family or observers, such as in Wii Fit ($n=1$), can create a feeling of uneasiness and provoke players to rather play in private. Lastly, three participants noted two smaller concerns: Two felt betrayed after learning about privacy concerns in Pokémon Go, and one experienced language barriers in Just Game, leading them to search for local gaming meetups.

5.8 Game Design Barriers

The smallest category is ● game design barriers with 48 mentions of 9 different problems (see Table 2). The 3 most common barriers are a frustrating game design, excessive difficulty, and a lack of variety. Since those barriers cannot be mitigated by players, these typically cause frustration and might lead to players quitting or switching the game.

The most stated flaw is a frustrating game design ($n=17$). Experienced with games like Pokémon Go ($n=8$), Wii Fit ($n=3$), and Beat Saber ($n=2$), participants report feeling frustrated, which may lead to them quitting ($n=3$), playing less ($n=2$) or feeling discouraged ($n=2$) or even enraged ($n=1$). As mitigation strategies, some participants try to take breaks or combine it with other activities (each $n=2$). Others practice more, encourage themselves, or use breathing exercises (each $n=1$). In games like Dance Dance Revolution ($n=3$), Just Dance ($n=2$), Beat Saber ($n=2$), participants also feel overwhelmed, exhausted (each $n=3$), frustrated, discouraged, or sad (each $n=1$) by experiencing excessive difficulty ($n=11$). Compared to frustrating game designs, this barrier can mostly be resolved by adjusting the difficulty ($n=6$), practicing more ($n=5$), or taking breaks ($n=2$).

Another problem relating to the game design is a lack of variety leading to repetition ($n=7$). Affecting games like Just Dance ($n=2$), Pokémon Go or Zumba Burn It Up ($n=1$), this can lead to boredom ($n=3$), reduce enjoyment ($n=2$) and contribute to quitting the game ($n=2$). To introduce more variety, participants typically use additional content ($n=3$), change game modes ($n=2$), or try other games ($n=1$). Similarly, some participants criticized paywalls ($n=5$) in Pokémon Go ($n=3$) and Just Dance ($n=2$). These are seen generally critical, causing frustration ($n=1$) or giving up on the game ($n=1$). Only one participant spends additional money ($n=1$), while another reported avoiding the additional content ($n=1$). Lastly, few participants reported smaller issues: Two people feel frustrated about perceived unfairness in the game design of Pokémon Go. One person wished games like Beat Saber or Half-Life Alyx would allow for more physical movements. Similarly, one participant had problems with user-generated maps in Beat Saber, and two participants disliked either that no game caters to their genre preference or that Pokémon Go did not meet their aesthetic wishes.

5.9 Platform-Related Differences

Across all 5 platforms and 6 types of barriers, participants reported 621 incidences of encountered problems. These spread almost equally among the 5 platforms, according to their respective prevalence: console ($n=317$), smartphone ($n=173$), VR ($n=92$), arcade ($n=30$), and other ($n=9$). This indicates that for each platform, 65-70% of participants who play games on that platform at least reported one barrier. While platforms appear similar at first glance, the differences become more apparent when splitting barriers by category (see Figure 5). For most platforms, environmental, physical, and technical barriers are most prominent and take up to two-thirds of the reported barriers. For example, the most commonly used platform, consoles, primarily suffers from environmental ($n=83$, 26.2%) and physical barriers ($n=82$, 25.9%), followed by technical issues

Table 2: The 29 reported barriers in the mental, social, and game design categories. The index numbers indicates their prevalence. The mentions column lists previous work that mentioned the barriers as motivation, discussion point, or potential issue. The findings/solutions column lists study results or developed solutions that help understand or resolve the barriers.

BARRIERS	MENTIONS	FINDINGS/SOLUTIONS
● Mental Barriers		
#4 addiction ($n=30$)	-	-
#6 in-game failure ($n=27$)	[23]	beating level after a long time creates sense of accomplishment [23]
#16 lack of motivation/commitment ($n=8$)	[104]	-
#18 cognitive involvement ($n=6$)	-	-
#18 unmet expectations ($n=6$)	[1, 50, 88, 89]	disappointment [88] and unmet health expectations [1] major predictors for quitting
#19 competitiveness/perfectionism ($n=5$)	-	-
#20 self-devaluationization ($n=4$)	-	-
#22 overthinking ($n=2$)	-	-
#23 collision anxiety ($n=1$)	[90, 91]	collision anxiety decreases with experience but application/environment have key influence [90]
● Social Barriers		
#3 playing game in-front of others ($n=36$)	-	-
#14 isolation ($n=11$)	[43, 50, 89]	group players need feeling of belonging [43, 89]
#15 arguments ($n=9$)	-	-
#16 lack of players ($n=8$)	[43, 88]	playing multiplayer exergames alone leads to worse adherence than single-player versions [43]
#17 clash with other obligations ($n=7$)	[88]	-
#20 considering other people ($n=4$)	[73]	players often do not use AR to avoid pointing phone at other people [73]
#21 lack of social support ($n=3$)	-	-
#21 harassment ($n=3$)	-	-
#22 privacy concerns ($n=2$)	-	-
#23 teasing ($n=1$)	-	-
#23 cultural/language barriers ($n=1$)	-	-
● Game Design Barriers		
#11 frustrating game design ($n=17$)	[5, 13, 38, 73]	exergames often too fast-paced/complex for older adults [5, 38], AR features uninteresting/annoying for many players [73]
#14 excessive difficulty ($n=11$)	[13, 15, 88, 89]	trouble reaching next level is reason to quit [88], people with mobility impairments have difficulties in not-tailored exergames [15]
#17 lack of variety/repetition ($n=7$)	[13, 41, 88, 89]	boredom and lack of variety main factor for decreased appeal [13, 41] or quitting [88]
#19 paywall ($n=5$)	-	-
#21 game design limits movement ($n=3$)	-	-
#22 perceived unfairness ($n=2$)	[23]	unfair levels create lingering frustration [23]
#23 problems with user-made content ($n=1$)	-	-
#23 dislike genre ($n=1$)	[50, 104]	-
#23 dislike aesthetics ($n=1$)	-	-

($n=82$, 16.7%). Notable exceptions from this pattern include the high importance of mental barriers for smartphone ($n=30$, 17.3%) and arcade games ($n=5$, 16.7%) or the dominance of physical problems for VR experiences ($n=27$, 28.1%).

Another interesting difference becomes apparent when looking at the most commonly noted barriers across categories. Over all device platforms, the “ranking” is unanimously led by the concern of limited space ($n=61$). The next barriers follow with over 40% fewer incidences: participation in a movement-intensive game ($n=37$), playing the game in-front of others ($n=36$), addiction ($n=30$),

collision ($n=28$), hardware-failures ($n=27$), in-game failure ($n=27$), injuries ($n=25$), fatigue ($n=22$), and pain ($n=20$). This top-10 list consists of 4 physical, 2 environmental, 2 mental, 1 technical, 1 social, and 0 game design barriers. While some barriers such as limited space and collisions were frequently reported, others like harassment or addiction, though less common, had more severe emotional and behavioral consequences. This distinction is crucial for prioritizing design interventions. Additionally, the prevalence of barriers also heavily depends on the device platform the exergame is played on. The top-3 noted barriers for each platform are:

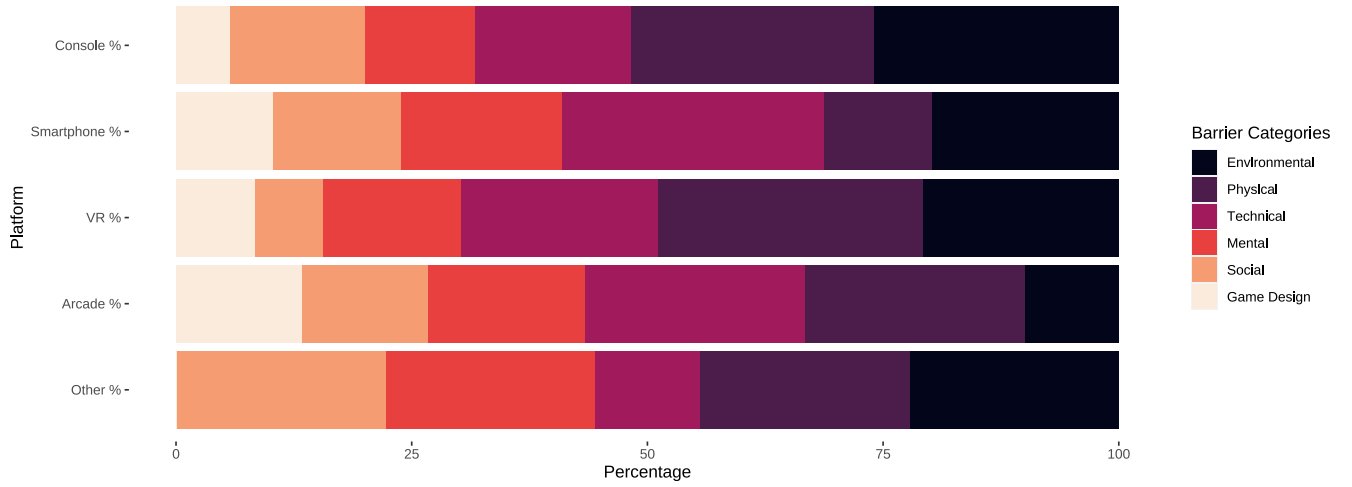


Figure 5: Distribution of encountered barriers across the six categories for all five device types: console, smartphone, VR, arcade, other (from top to bottom). Generally, the environmental, physical, and technical barriers make up the biggest chunk of barriers for almost all platforms.

- console: limited space ($n=50$), participation in movement-intensive game ($n=25$), playing the game in-front of others ($n=20$)
- smartphone: addiction ($n=18$), geological limitations ($n=16$), GPS issues ($n=15$)
- VR: collisions ($n=9$), limited space ($n=8$), pain ($n=8$)
- arcade: playing the game in-front of others ($n=3$), maintenance ($n=2$), being overly competitive ($n=2$)
- other: participation in a movement-intensive game ($n=2$), isolation ($n=1$), internet connection ($n=1$)

5.10 Cross-Category Interactions

While our analysis categorized barriers into six distinct types, many issues reported by participants reveal cross-category interactions that highlight the compound nature of exergame challenges. For example, technological barriers such as hardware-failures ($n=40$) and tracking problems ($n=53$) are often linked to collisions, which in turn can cause injuries ($n=36$) or lead to players failing the game ($n=37$). The latter often has a social consequence as well, since players report subsequent embarrassment when performing in front of others ($n=31$). Another frequent observation was the interplay between excessive in-game difficulty, fatigue ($n=15$), and in-game failures ($n=35$), forming a cognitive–physical–emotional spiral that undermines motivation. On the opposite end of the motivational spectrum, over-engagement in the form of addiction ($n=30$) or lacking fitness ($n=8$) can cause injuries as well. Our data also shows potential origins for social tensions: being overly competitive and perfectionistic was frequently linked to player-to-player collisions ($n=7$). These collisions, in-game bugs, as well as pets and bystanders as additional collision sources often go hand-in-hand with subsequent arguments ($n=32$), showing the clash between gaming activities and the outside world. Lastly, we found the impact of personal taste on the game perception, since participants who disliked the genre often also complained about the aesthetics ($n=6$)

and a lack of variety ($n=10$). These patterns highlight the need for holistic design approaches that address not only individual barriers but also their interdependencies.

6 Discussion

With the creation and adoption of commercial exergames on the rise, our research has focused on understanding the barriers faced when playing these games. We discuss our findings below by addressing our research questions and contextualizing our results within the landscape of exergame research and development. Finally, we offer a comprehensive analysis of the implications of our findings, providing insights and recommendations to guide future research and development in the field of exergames.

6.1 RQ1: What are common barriers encountered while playing commercial exergames?

Our survey revealed 60 distinct barriers, reported over 600 times, with a strong tendency toward a subset of highly prevalent issues. Specifically, 21 barriers were mentioned at least 10 times and accounted for 77.1% of all reported incidences. As seen in Table 1 and Table 2, about two thirds of reported barriers have been at least mentioned once in prior literature, showcasing the confirmatory and organizing nature of our research. At the same time, most papers only mentioned barriers anecdotally or motivational; the *interesting findings* column, collecting results from studies, surveys, or designs, is mostly empty. This highlights the need for a deeper exploration and resolution of individual barriers in future work. Focusing on the most commonly reported barriers, we can group them into three overarching types based on their impact on the play experience:

Interruptive Barriers. A third of barriers preclude or disrupt the game experience due to external factors: limited space, collisions, hardware-failures, internet connection problems, geological limitations, pets and people, GPS, and device support. Some of the raised issues, in particular planned hardware obsolescence or excessive battery drain, have surfaced as primary reasons for permanent disengagement. Nonetheless, they fall under the shared responsibility of hardware manufacturers and game developers, making them hard to resolve. The remaining problems are specific to exergames and highlight their situated nature [97]: Play unfolds in real-world environments that are often unpredictable and shared with others. Unlike other genres, exergames require physical playspace free of collision sources and reliable tracking within. While meeting these requirements is mostly up to the players, developers should assist them by designing for spatial flexibility (e.g., dynamically adaptive game tasks) and unforeseen interruptions (e.g., auto pause). This would allow more players to enjoy exergaming despite environmental constraints.

Physical and Mental Strain. The second main type occurs during normal gameplay but has a lasting impact beyond the play session: exhaustion, addiction, injury, fatigue, pain, overstraining, and isolation. These issues show how exergames extend the original idea of embodied interactions [28] by blurring the boundaries between the physical and digital divide. The players' bodies do not only become a central part of the game and the primary digital input source to control it, but also a source of vulnerability that can impact the real world. Accordingly, designing for movement intensity is a double-edged sword. While bodily movements are the core feature of exergames, they also frequently cause exhaustion and injuries. In particular, VR games with their view-blocking headsets are prone to cause injuries from collisions [24] and even create collision anxiety [90]. Regarding mental barriers, the frequent mentions of addiction and isolation hint towards limited autonomy control of players towards these games and unsupported relatedness needs when viewed through the lens of Self-Determination Theory [25]. Besides more opportunities for engaging with other exergame players, the issues also highlight the need for proper pacing mechanisms in the game design, as previously discussed by Crane et al. [23]. Automatic exercise adjustments, breaks, and longer pauses that are deeply integrated into gameplay would help prevent both physical overstraining and excessive engagement (i.e., addiction and perfectionism). For instance, Schneider and Graham [92] showed the effectiveness of nudges to keep players in the optimal exertion range.

Enjoyment-Related Barriers. These barriers affect the emotional and social experience of play and *prevent players from enjoying the game fully*: playing in front of others, in-game failures, tracking problems, frustrating game design, bugs, and excessive difficulty. This list of factors complements Crane et al. [23]'s list of frustrating sources and, interestingly, does not encompass problems relating to interface complexity, as reported by Harrington et al. [38]. Instead, we found that most players emphasized the impact of social and mental factors on the game enjoyment. The positive effect of social play for adherence has been both explored through prior user studies [43] and discussed with the help of established frameworks [14]. However, while the social aspect can be an important source for motivation, it could also evoke embarrassment or

even harassment for both bystanders and co-players. For instance, Arjoranta et al. [3] reported cases where the lowered social barrier of exergames caused uncomfortable interactions with strangers. The more severe form, toxic behavior, is unfortunately also common in multiplayer experiences beyond exergames [52, 53] and can hardly be avoided. These issues might be particularly concerning because they erode the intrinsic motivation that exergames typically rely on. To mitigate negative effects as much as possible, developers should focus on alternative game modes and targeted designs that build confidence and motivation, and allow players to connect with like-minded peers more easily. Alternatively, we could also go the opposite way and embrace the embarrassment of performing in exergames, for example, by using unusual movements that spark enjoyment through looking silly for both players (similar to designing purposefully shitty user experiences [21]).

6.2 RQ2: How do barriers in exergames differ between different technologies?

Our analysis revealed platform-specific differences in both reported barriers and play frequency. Console games were the most widely played, while smartphone games were played most frequently (48% of players reported playing daily). All other categories only played a minor role. This observation matches public attention. In the early 2000s, affordable motion-tracking consoles like the Wii and Xbox Kinect let many players first experience exergames like Wii Fit—creating reminiscence and nostalgia [9]. This hype was ultimately repeated with the launch of Pokémon Go as the first, and so far only well-performing, smartphone-based exergame. Its incentivizing push notifications and constant company on the personal smartphone likely explain the higher usage frequency.

In terms of barriers, console games were most affected by spatial constraints and social discomfort, particularly embarrassment of playing in front of others. These findings extend the work of Harrington et al. [38] on usability challenges for older Kinect users and likely stem from the typical living room location, which is often co-populated by other family members and pets. The social challenges are even more pronounced for arcade games, which are almost always played in a public space. VR games, especially when using mobile headsets, provide more flexibility to choose a private space but also come with different challenges, including collisions with unseen obstacles, pain stemming from repetitive wrist movements, or, as also reported by Shaw et al. [94], cybersickness, and display problems. Lastly, smartphone games can avoid all of these problems but suffer from unique location-based issues, such as GPS inaccuracies, geographical and weather constraints [63], and addictive play patterns (e.g., through free-to-play models). This observation is in line with work by Laato et al. [56], who found that the environmental context and the player community shape motivation and engagement with Pokémon Go.

Overall, our findings suggest that barriers are deeply rooted within the situated use context of each platform. While prior studies have examined barriers within specific technologies, our cross-platform comparison reveals how these issues shift depending on the device, environment, and social setting. Accordingly, we see the need for platform-sensitive design strategies that tailor games to the unique surrounding context of each device category.

6.3 RQ3: What are the effects of encountered barriers on the exergaming experience?

Our data revealed that while most barriers have tangible or emotional consequences, participants often did not articulate these effects in detail. Out of 621 barrier reports, only 313 were accompanied by explicit outcomes. For the reported effects, the observations show that, despite a variety of 43 effects, most barriers influence players similarly. Many participants state negative feelings like frustration ($n=63$), exhaustion ($n=51$), and embarrassment ($n=32$). Players might also reduce their playtime ($n=29$) or quit the game ($n=35$). Together, these notions account for 67% of reported effects. Our findings align with prior work on exergame disengagement. Crane et al. [23] identified frustration, fatigue, and pain as barriers to participation for older adults, while Rasche et al. [88] emphasized boredom and disappointment as reasons to quit. We extend these insights by showing that such effects are not limited to specific demographics or platforms but are widespread across commercial exergames.

While physical barriers like injuries and fatigue can lead to frustration, most players seem to accept them as natural concomitants of exergaming and focus on adaptation strategies such as using protective gear. In contrast, social barriers, most profoundly negative reactions from other people or the fear thereof, are not easily mitigated and lead to deep feelings of embarrassment and lowered self-esteem, ultimately leading to disengagement. These findings align with Kordyaka et al. [52], who linked continuous exposure to toxic behavior with lowered self-esteem in multiplayer games. Even if players do not participate in multiplayer activities, psychological factors such as addiction, frustration from failure, and perfectionism still affect long-term engagement. Rasche et al. [88] and Adam and Senner [1] found that disappointment and unmet health expectations are major predictors for quitting exergaming. Both categories highlight the need to prioritize encouraging over captivating gameplay, self-paced progression over predetermined difficulties, and confidence-building social features over limitless interactions between players. Compared to the more emotional consequences of mental or social barriers, spatial issues typically cause immediate and tangible risk factors. Console and VR games require considerable space in living rooms, and the lack thereof is not only detrimental to performance but also creates safety risks. Even with enough space, energetic in-game movements can cause unintended positional drifts that might lead to collisions, as stated by Cmentowski et al. [20]. Lastly, for location-based games, these safety risks are external; the flow of the game can lead to a disconnection from the real world, raising the likelihood of dangerous encounters with other road users.

In summary, our findings suggest that the effects of barriers are not solely determined by their category but by their visibility, recurrence, and social context. While physical discomfort may be tolerated, social embarrassment and repeated failure are more likely to deplete motivation and lead to disengagement. Future research should focus on confidence-building designs, balancing challenge with accomplishment and social experiences with private play.

6.4 RQ4: How do players overcome/mitigate encountered barriers?

Our data shows that players are generally proactive in resolving barriers, often relying on ad-hoc, self-devised strategies. We identified 52 distinct mitigation strategies ($n=52$). For example, players would rearrange their furniture to make space, run cables across the apartment, or lock pets into other rooms—even if these measures take additional setup time. Some prior work has attempted to assist players in this process, for example, by creating pet detection algorithms [18]. The problem-solving mentality also reflects remarks by Laato et al. [56], who noted that Pokémon Go players formed car-sharing communities to reach distant locations, thereby overcoming environmental constraints. However, our findings also reveal a clear divide between barriers that can be mitigated through individual action and those that require systemic design interventions. For example, physical barriers such as fatigue or pain can be avoided to a certain degree by incorporating regular breaks, stretching, and breathing exercises into the routines.

Unfortunately, this problem-solving mentality cannot resolve many of the mental and social barriers, like addiction or harassment. Being unable to find ideal solutions except for playing less or quitting completely, participants often reported feelings of frustration and attempts to ignore the problem. Alexandrovsky et al. [2] notes that frustration and unresolved problems are often reasons for disengagement, which in turn can cause rage or regret when players feel like they wasted precious time. Notably, some participants attempted to reframe their experience, for example, by “*focus[ing] on fun*” (P174) rather than performance or playing only in private settings. This suggests a form of situated resilience, where players negotiate their environment and expectations to maintain engagement. Crane et al. [23] also found that persistence can transform barriers, for example, when feeling accomplished from beating a level after repeated failure. However, such strategies are fragile, especially when barriers remain or are socially charged.

In sum, while players demonstrate resilience and creativity in mitigating barriers, their strategies are often reactive and external to the game system. This highlights the need for designing exergames that not only detect and adapt to barriers but also guide players through mitigation, especially for mental and social challenges that are harder to self-manage. Potential safeguarding features include pacing mechanisms to mitigate addictive patterns, social filtering against harmful connections, and dynamic game tasks that adapt to a variety of room layouts.

6.5 Contextualization within Exergame Landscape

We based our organization of exergame barriers solely on experiences reported by players since this allows for a comprehensive view of pain points identified by the current user base. However, we do not claim completeness; notably, barriers that prevent people from starting to play in the first place (e.g., accessibility [5, 35]) or hinder organizations from using exergames for health interventions [75] are not captured with our approach. Similarly, Nelson et al. [73] found that the AR functionalities featured by most current smartphone games can be another source of barrier not reported in our study. Participants in their survey found AR capabilities just

a mere gimmick and refrained from using them to avoid pointing their phones at other people. Furthermore, we have to assume that most of the issues are known to developers, given that they have been reported in prior literature. This raises the question of why these barriers persist in many of today's exergames. We attribute this to a general inertia in game adoption (i.e., many of the listed games were developed more than 10 years ago) and to a conflict between improved accessibility and higher costs that makes resolving barriers economically unfeasible for developers. However, we believe that increased visibility and a systematic organization of barrier reports is an essential step towards more accessible and usable exergame experiences.

Not all of the 60 identified barriers have the same importance or the same impact on players. While some cause severe physical harm or trigger players to quit, others are mostly ignored or easily circumvented. This raises the question about how to prioritize barriers, especially when investing limited resources to exploring and resolving them. An important aspect that distinguishes exergames from other entertainment media is the concept of retention. Chan et al. [14] discuss that retention can not only refer to participation in a single game but to retaining an active lifestyle in general: "players may discontinue playing a game, but continue being active through playing other exergames or participating in other physical activities/sports". Based on this idea, we think that the most critical barriers are those that impede retention within the exergame genre or general physical activity and that have a significant impact on players' lives and well-being. In particular physical and environmental barriers with safety implications (e.g., distractions, unsafe playspaces, collisions, and injuries) as well as mental and social barriers that harm players' mental well-being (e.g., addiction, anxiety, feeling of isolation, harassment) are essential to be addressed head-on. Technological and game design barriers mainly cause frustration and are typically more confined to single games.

6.6 Implications for Exergame Research and Development

Based on the barriers and mitigation strategies identified in the previous sections, we propose five design implications that challenge common assumptions and offer concrete directions for future exergame development.

6.6.1 Design for Shared Spaces. Exergames are often motivated by the increasing physical inactivity and framed as playful alternatives to traditional workouts. However, unlike gym-based exercise, they are typically played in shared, domestic, or public spaces. These environments are not optimized for movement and introduce risks such as collisions with furniture, pets, or bystanders. Compared to other game genres, exergaming also requires much more space to perform bodily movements, which clashes with the outside world more frequently and might induce collision anxiety among players [90]. Our findings show that players try to mitigate problematic situations by creating more secured space, e.g., by rearranging furniture. Nevertheless, this is not a sustainable solution and, depending on the circumstances, not sufficient.

We argue that exergames should adapt to the player's environment, not the other way around. For example, games like Just Dance could remix dance choreographies to **fit the momentarily**

available space, choosing between more stationary and wider movements automatically. To our knowledge, this concept has not been explored in prior work but aligns well with recent advances in collision anxiety research [90]. While such adaptations might decrease the exercise effectiveness of exergames, our data suggests that enjoyment and safety are more critical for long-term engagement than maximizing exertion. This challenges prior assumptions on why players quit exergaming [1]: Only six participants complained about unmet expectations, primarily criticizing the focus on difficulty rather than enjoyment. For example, one person found Audica [100] to be "*more focused on difficulty rather than following a rhythm*" (P28). Accordingly, we also emphasize the intricate balance between prioritizing enjoyment and exercising in exergames.

6.6.2 Design for Fatigue Awareness. Many of the physical barriers we identified—overstraining, pain, injuries, or even fractures—stem from overuse and lack of pacing. Participants explained becoming too immersed, competitive, or perfectionistic to notice their physical limits. These issues contradict the intended positive impact of exergaming and could ultimately seriously affect players' health by causing overexertion. However, they could also be easily addressed in game design through **personalized routines** (e.g., [110]). Building on the well-researched dynamic adaptation of the in-game difficulty, we propose the integration of fatigue-aware game designs, such as:

- automatic break suggestions based on session length or movement intensity
- visualizations of harmful movements and excessive strain (e.g., wrist usage in Beat Saber)
- adaptive cooldowns or "wind-down" modes after intense play

While there has been considerable research into steering players towards optimal exertion [92] and into adapting game difficulty based on in-game performance [20] or physical effort [95], we are yet to see an integrated coaching system that incorporates all proposed elements. Additionally, we see the need for future research into positive disengagement experiences to allow for feelings of "satisfaction and closure" [2] rather than addiction or frustration.

6.6.3 Design for Adoption and Accessibility. Our data does not allow for a clear differentiation between past and present game experiences, as participants were asked to report on both currently and previously played games. However, the high prevalence of 15- to 20-year-old console games in our dataset indicates still strong reminiscence and possibly even continued use. In comparison, VR exergames still remain niche. Rather than interpreting this as a failure of XR exergames, given that XR devices are not affordable for everyone, we propose that future design should focus on each platform's unique characteristics. Console games, played in shared living rooms, are ideal for co-located play and social visibility. In contrast, VR headsets have the power to take players out of their usual surroundings, providing a distraction-free exergame experience, at the risk of collisions or feelings of being observed. An interesting case is the prevalence of smartphone exergames: Pokémon Go heavily dominated the field with 97% of smartphone players. This focus on single applications with high media coverage shows that new exergames have a hard time standing on the market. To

open consumers to new experiences and platforms, we see the need to **ease access** to exergames, echoing recommendations by Mueller et al. [70]. For example, advancing device interoperability to enable reusing existing devices and focusing on the supportive aspect of co-located play can lower the existing entry barrier.

6.6.4 Design for Novelty, Personalization, and Balance. Although repetition was not a very common barrier, it was disproportionately mentioned by experienced exergame players. These users had already exhausted the available content in games like Just Dance or Beat Saber, leading to boredom and disengagement, thereby preventing long-term retention. This decline in engagement has already been noted multiple times [13, 41, 88] and can be prevented or reversed through regular feature updates, as found by Zhao et al. [111]. Therefore, we see great potential in integrating **procedural content generation** as an easy and effective way of preserving a sense of novelty. Furthermore, the recent advancements in artificial intelligence (AI) could help to make the exergaming experience more personalized and beneficial. First studies are assessing the impact of AI-powered exergame elements, for example for assessing exercise performance [85, 86] or creating virtual assistants [55]. However, we believe that the true potential goes far beyond these efforts: Apart from generating tasks that fit the individual fitness level, generative AI could help in **guiding players** to perform more complex movements optimally (cf. [48]), thereby reducing the risk of injuries. In multiplayer experiences, machine learning can improve balancing in an unobtrusive way to avoid the impression of unfairness or a decrease in self-esteem (cf. [35]). Lastly, it can empower players to generate new content for games, avoiding frustration with poorly designed levels: *“I found a beatmap of a song I like and would love to play in the game but the creator has not done a good job at all” (P30).*

6.6.5 Design for Confidence and Safety. Social features are a double-edged sword. While peer support can boost motivation, many participants felt uncomfortable playing in front of other people. As a first important addition towards more exergame engagement, we therefore see a need for the inclusion of **confidence-building features**, such as beginner-friendly modes, empowering progression systems, and supportive guidance. These concepts are similar to prior work on guiding movements in exergames [48], encouraging self-competition [66], or motivating players by giving them more muscular avatars [51]. Some participants also felt embarrassed playing exergames that were perceived, in their eyes, childish by others. To counter this impression, we should think about creative ways of **integrating exergames into daily life**, for example, by creating a cleaning-inspired exergames similar to early prototypes [26]. Such hybrid experiences would also make it easy to combine daily life with exergame engagement. Unfortunately, we also observed a dark side of social features: harassment. Shared by most multiplayer genres, such behavior is intolerable and calls for further research to identify optimal mitigation strategies. As a starting point, we advocate for providing players with **more control over social interactions** and with **community-building tools** to connect with like-minded players safely.

6.7 Limitations

Like any empirical work, our study is not without limitations. Firstly, our collected qualitative data may have been affected by recall biases and social desirability effects. For example, participants may have underreported sensitive issues such as harassment or addiction. Similarly, using the online recruitment platform Prolific creates the risk of self-selection bias, potentially skewing the sample toward more tech-savvy or motivated players. However, to understand the barriers, we need to focus on studying people who actually play these games. Secondly, while we aimed for a wide dataset querying both past and present exergame experiences, this study design also limits our ability to assess current platform adoption or longitudinal engagement patterns. Accordingly, we recommend future research to conduct small-sample qualitative methods to explore the identified barriers and their resolution in more detail Müller et al. [72]. Thirdly, while we queried effects and mitigation strategies, we do not specifically discuss the barriers’ severity or cumulative impact on users’ well-being and health. Some issues may be tolerable in isolation but become problematic when compounded. Therefore, we recommend future work to explore barrier interactions and longitudinal implications. Lastly, despite being rigorous, our coding was ultimately interpretive and shaped by the authors’ backgrounds in exergame research. While we do not view this as a bias but rather as an asset in understanding the exergames and their associated challenges, our limited expertise in medical and movement science research should be considered a constraint limiting in-depth interpretation about certain issues, such as the mental barriers category.

Finally, while our research mainly focused on identifying barriers, it also highlights that commercial exergames are indeed enjoyed by a wide and diverse player base despite their shortcomings. We believe these games offer valuable insights into what works well when designing for different user groups with varying needs. Therefore, our goal is not only to highlight the issues players encounter but also to support the continued use of these exergames.

7 Conclusion

This work explored which barriers are commonly experienced by players of commercial exergames, how these affect the game experience, and how players attempt to mitigate them. Our online survey with 174 participants yielded a rich dataset of 60 distinct barriers across 6 categories, 43 effects, and 52 mitigation strategies. Our findings show that barriers interrupt the gameplay through external factors, like collisions or hardware-failures, decrease enjoyment during exergaming, e.g., due to embarrassment, failure, or bugs, and even affect players’ everyday life beyond the exergame session, e.g., in the form of injuries or addiction. Despite individual differences, most barriers are shared between device types. While many players respond with creative ad-hoc solutions, mental and social barriers often remain unresolved and can lead to frustration or disengagement.

Based on these findings, we proposed five design implications and directions for future research. In particular, we envision games that adapt automatically to the momentarily available space to minimize interference with the public environment exergames are played in. We believe in the potential of personalized routines that aid with disengaging after reaching the exercise goals to prevent

overexhaustion. We discuss how to embrace platform-specific differences to increase adoption of new technology. Finally, we propose the addition of procedural content and confidence-building features to mitigate repetition effects and social problems. Taken together, this work advances our understanding of the “dark sides” of exergaming and offers actionable insights for designing inclusive, adaptive, and emotionally intelligent movement-based games.

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